

Plaque Radiotherapy for Juxtapapillary Choroidal Melanoma Overhanging the Optic Disc in 141 Consecutive Patients

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Objective: To evaluate tumor control with plaque radiotherapy for juxtapapillary choroidal melanoma that overhangs the optic disc.

Methods: Retrospective medical record review of 141 consecutive patients with data on complications of treatment, final visual acuity, visual loss, enucleation, tumor recurrence, metastasis, and death.

Results: The median patient age was 61 years. Presenting symptoms included reduced visual acuity in 72 eyes (51%), photopsia in 14 (10%), and visual field defect in 18 (13%); 35 patients (25%) were asymptomatic. The median tumor basal diameter was 11 mm and the median thickness was 5.2 mm. The tumor overhung 50% or less of the disc in 88 eyes (62%) and more than 50% of the disc in 53 eyes (38%). In 19 cases (13%), the tumor overhung the entire disc. All patients were treated with plaque radiotherapy, using a notched design in 126 eyes (89%) and a round design in 14 eyes (10%), with iodine 125 in 132 eyes (94%) and cobalt 60 in 9 eyes (6%). The me-

dian radiation dose to the tumor apex was 8500 cGy. Adjuvant transpupillary thermotherapy was used in 54 eyes (39%). During a mean follow-up of 56 months, complications included nonproliferative retinopathy in 61 eyes (51%), proliferative retinopathy in 26 (22%), maculopathy in 44 (37%), papillopathy in 57 (48%), neovascular glaucoma in 23 (19%), and vitreous hemorrhage in 48 (40%). A final visual acuity of 20/200 or worse was measured in 72 eyes (77%), and visual loss of more than 5 Snellen lines occurred in 59 eyes (63%). Enucleation was necessary in 27 eyes (23%). Tumor recurrence was found in 12 eyes (10%). Metastasis developed in 15 patients (13%) and death in 4 cases (3%).

Conclusions: Using plaque radiotherapy for choroidal melanoma overhanging the optic disc, local tumor control was achieved in 90% of cases. Tumor and radiation effects led to poor visual acuity in 77% of eyes. The metastatic rate was 13% and the mortality rate was 3%.

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THE CONSERVATIVE TREATMENT of choroidal melanoma adjacent to the optic disc (juxtapapillary choroidal melanoma) poses a unique challenge owing to the posterior tumor location within the globe, the necessary precision in tumor localization, the difficult dissection of periocular tissues near the optic nerve, and the complexity of radiation field design.¹⁻³ Juxtapapillary choroidal melanoma has been treated by a range of modalities, including plaque radiotherapy,^{4,5} proton beam radiotherapy,⁶ stereotactic radiotherapy,⁷ transpupillary thermotherapy,^{8,9} and enucleation.¹⁰ A previous report from our group showed that plaque radiotherapy for juxtapapillary melanoma showed similar patient survival to enucleation.⁵ The Collaborative Ocular Melanoma Study (COMS) evaluated plaque radiotherapy vs enucleation for choroidal melanoma, but

did not consider juxtapapillary tumors.¹¹

In some instances, juxtapapillary choroidal melanoma herniates through or around Bruch membrane into a mushroom configuration and partially or completely overhangs the optic disc. This special group of patients is usually treated with enucleation, but, in certain circumstances, plaque radiotherapy is used, especially if there is poor or no vision in the fellow eye. In these cases, the radiation dosimetry must be calculated to extend as a solid ellipse posterior to the plaque applicator in a field to incorporate the entire tumor while at the same time minimizing the radiation dose to the sclera, optic nerve, and macular region. The custom design to the radiation field over the optic disc is perhaps the most challenging of ocular treatment plans and is most effectively achieved in a notched plaque. In this retrospective case series, we describe our experience with custom-designed plaque

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radiotherapy for juxtapapillary choroidal melanoma that overhang the optic disc.

METHODS

We searched the electronic database of patients seen at the Oncology Service of Wills Eye Institute, Philadelphia, Pennsylvania, for juxtapapillary choroidal melanoma treated with plaque radiotherapy. Juxtapapillary choroidal melanoma was defined as a choroidal melanoma with its posterior margin located within 1 mm of the optic disc. Eyes in which the apical portion of the melanoma overhung and partially or completely obscured the optic disc were included in this analysis. The study comprised a retrospective review of the medical records of 141 consecutive patients who were evaluated from April 1980 to November 2004. Two patients failed to complete more than 3 months of follow-up and were excluded from the outcomes analysis.

Cases were stratified according to percentage overhang of the optic disc (**Figure**). The amount of disc overhang, defined as percentage of the optic disc surface blocked by the overhanging melanoma, was determined by indirect ophthalmoscopy with a 20-diopter lens by 1 of the 2 senior authors (C.L.S. or J.A.S.). Tumor thickness was measured by B-scan ultrasonography. All tumors were evaluated for circumferential involvement (total number of clock-hours of tumor involvement adjacent to the optic disc), and those tumors with 100% overhang of the optic disc were specifically evaluated by B-scan ultrasonography for the extent of circumferential involvement.

Our methods of clinical examination, diagnosis, and treatment have been described previously.¹ In this study, we recorded patient details at presentation, including age, race, sex, medical history (hypertension, diabetes mellitus, hypercholesterolemia, dysplastic nevus syndrome, and skin melanoma), and family history (choroidal or skin melanoma). The general ocular details included patient symptoms, visual acuity, anterior segment details, and intraocular pressure. Specific tumor details included quadrant of tumor epicenter, number of clock-hours the tumor encompassed the optic nerve, distance in millimeters to the optic disc, percentage overhang of the optic disc by the tumor, distance to the foveola in millimeters, largest basal diameter in millimeters, thickness in millimeters, tumor shape, growth pattern (dome, mushroom, or diffuse), color (pigmented, partially pigmented, or nonpigmented), extraocular extension, retinal invasion, subretinal fluid, orange pigment, optic disc swelling, and retinal vascular congestion. The plaque radiotherapy details included information on radioactive isotope (iodine 125 or cobalt 60), shape (round, notched, or rectangular), distribution (full or posterior), time of radiation exposure (hours), and radiation dosage and rates to the tumor apex, tumor base, optic disc, foveola, and lens. Additional treatment details included the number of sessions of adjunctive transpupillary thermotherapy (TTT) and treatment strategy, adjunctive argon laser photocoagulation or krypton laser photocoagulation. Patients who had TTT or adjunctive argon laser photocoagulation/krypton laser photocoagulation as a primary treatment before plaque radiotherapy and those who had adjunctive argon laser photocoagulation as planned adjunctive treatment following plaque radiotherapy were excluded from the outcomes analysis. Follow-up data included radiation complications (nonproliferative and proliferative radiation retinopathy, maculopathy, papillopathy, radiation cataract, neovascular glaucoma, vitreous hemorrhage, and scleral necrosis), thermotherapy complications (venous or arterial occlusion, thermal damage to optic nerve), final visual acuity of 20/200 or worse, visual loss of more than 5 Snellen lines, need for enucleation, and recurrence of tumor (any evidence of tumor regrowth). Out-

comes of systemic metastasis (judged by clinical examination, liver function serology, and chest and liver imaging) and deaths from metastatic disease were also recorded.

RESULTS

CLINICAL FEATURES

During the study period, 141 consecutive patients with juxtapapillary choroidal melanoma overhanging the optic disc were treated with custom-designed plaque radiotherapy. The patients' demographic features are presented in **Table 1**. The median age at presentation was 61 years and most patients were white (99%) and male (61%). The patients' presenting symptoms were reduced visual acuity in 72 eyes (51%), photopsia in 14 (10%), and visual field defect in 18 (13%). There were 35 asymptomatic patients (25%). Median visual acuity at presentation was 20/40.

The tumor characteristics are presented in **Table 2**. The median tumor basal diameter was 11 mm (mean, 11 mm; range, 4-20 mm) and median thickness was 5.2 mm (mean, 5.7 mm; range, 1.3-14.8 mm). The shape of tumors on B-mode ultrasonography was plateau or dome in 127 eyes (91%) and mushroom in 13 (9%). The tumor base touched the optic nerve in all cases, and optic disc swelling was found in 5 eyes (4%). The extent of optic disc overhang was subdivided into 50% or less in 88 eyes (62%) and greater than 50% in 53 eyes (38%). There were 19 eyes with 100% overhang (13%). The circumferential extent of the tumor at the disc was less than 4 clock-hours in 16 eyes (11%), 4 to 8 clock-hours in 95 (67%), and greater than 8 clock-hours in 30 (21%).

In total, 27 eyes had small melanomas less than 3 mm in height. All 27 tumors showed 3 or more risk factors for growth.^{12,13} All 27 small melanomas were juxtapapillary, 26 (96%) were greater than 2 mm in elevation, 22 (81%) showed signs of subretinal fluid, 17 (63%) had overlying orange pigment, and 19 (70%) were symptomatic. A total of 3 risk factors were found in 6 eyes (22%), 4 risk factors were found in 12 eyes (44%), and all 5 risk factors were found in 9 eyes (33%).

TREATMENT CHARACTERISTICS

Plaque radiotherapy parameters are listed in **Table 3**. The plaque shape was notched in 126 eyes (89%), round in 14 eyes (10%), and rectangular in 1 eye (<1%). The median radiation dose was 8500 cGy to the apex and 32700 cGy to the base. The most commonly used radioisotope was iodine 125 in 132 eyes (94%), though, in earlier cases, 9 eyes received irradiation with cobalt 60 (6%). Radioactive seeds were positioned on the plaque in a custom-designed manner in an effort to spare normal tissues and reduce radiation dose to the optic nerve. The distribution of the radioactive seeds covered the entire plaque area (full distribution) in 104 eyes (76%) and was posterior-weighted (posterior distribution) in 33 eyes (24%). The results regarding planned adjunctive therapy are summarized in **Table 4**.

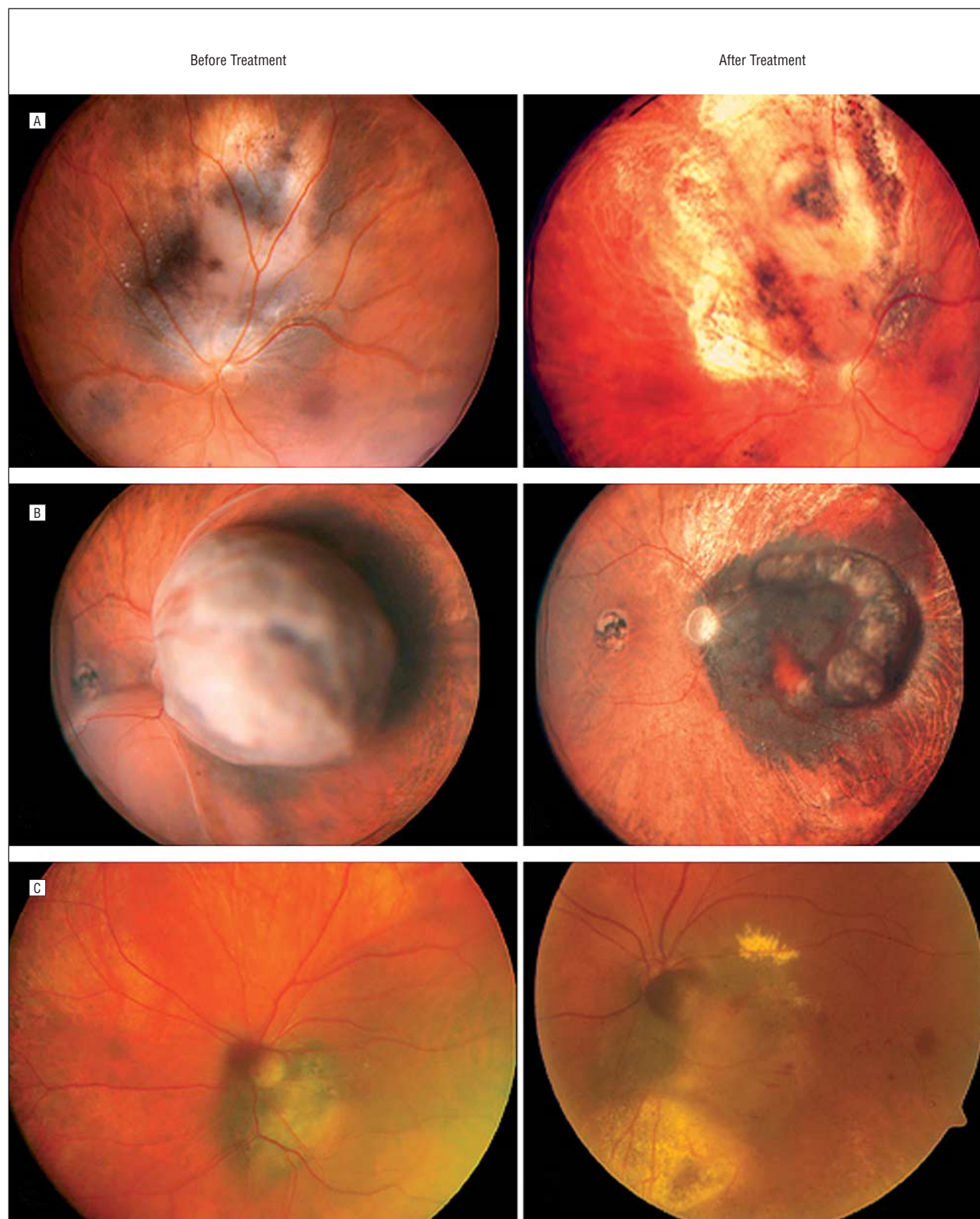


Figure. Treatment of juxtapapillary choroidal melanoma overhanging the nerve with custom-designed plaque radiotherapy. A, Fifty-eight-year-old white man with choroidal melanoma of 3.0 mm thickness and with less than 50% overhang of the optic disc. Visual acuity was 20/200. Custom-designed plaque radiotherapy and transpupillary thermotherapy (TTT) were applied. Sixteen months following treatment, tumor control was achieved and visual acuity was 20/40, with a tumor thickness of 2.6 mm. B, Fifty-four-year-old white man with a juxtapapillary melanoma displaying greater than 50% overhang of the optic disc and serous retinal detachment. Note chorioretinal scars attributed to previous presumed ocular histoplasmosis syndrome. Visual acuity was 20/70 and tumor thickness was 9.6 mm. Custom-designed plaque radiotherapy and TTT were applied. Four months following treatment, tumor regression was achieved and thickness reduced to 4.2 mm. Visual acuity was no light perception. C, Seventy-nine-year-old white woman with 100% overhang of the optic disc by a juxtapapillary choroidal melanoma with a thickness of 3.0 mm. Visual acuity was 20/40. Custom-designed plaque radiotherapy and TTT were applied. Twenty-six months following treatment, tumor regression was achieved, with a thickness of 2.0 mm; visual acuity was 20/200.

Table 1. Demographics of Patients With Choroidal Melanoma Overhanging the Optic Disc

Characteristic	Patients, No. (%) (n = 141)
Age, median, mean (range), y	61, 60 (24-87)
Race/ethnicity	
White	139 (99)
African American	0
Hispanic	2 (1)
Asian	0
Multiple	0
Sex	
M	86 (61)
F	55 (39)
Medical history ^a	
Hypertension	54 (38)
Diabetes mellitus	14 (10)
Hypercholesterolemia	11 (8)
Dysplastic nevus syndrome	0
Skin malignant melanoma	1 (<1)
Family history	
Choroidal melanoma	2 (1)
Skin melanoma	4 (3)
Neither	135 (96)
Ocular symptoms	
Decreased visual acuity	72 (51)
Flashes/floaters	14 (10)
Visual field defect	18 (13)
Pain	2 (1)
Diplopia	0
Asymptomatic	35 (25)

^aSome patients had more than 1 condition.

TREATMENT OUTCOMES

Treatment outcomes were analyzed for the whole group and subdivided into those who received adjuvant TTT (TTT group) and those who did not (non-TTT group). The entry criteria for the 2 groups were found to match, except for the greater rate of orange pigment ($P < .001$) and pigmented tumors ($P = .03$) in the TTT group and greater tumor thickness ($P < .001$) and tumor base ($P = .04$) in the non-TTT group (eTable 1, available at <http://www.archophthalmol.com>).

The outcomes of treatment are listed in eTable 2 and and **Table 5**. Nonproliferative retinopathy developed in a total of 61 eyes (51%); 26 eyes (22%) progressed to proliferative retinopathy. Maculopathy was found in 44 eyes (37%) and papillopathy developed in 57 eyes (48%). Neovascular glaucoma was observed in 24 eyes (20%), of which 19 eyes were in the non-TTT group (29%) and only 5 eyes were in the TTT group (9%). Of the 48 eyes (40%) with vitreous hemorrhage, 37 (77%) had only mild bleeding managed with observation for resolution, 2 (4%) required panretinal photocoagulation, and 6 (13%) required pars plana vitrectomy; in 3 eyes (6%), the treated tumor could not be monitored satisfactorily in the presence of vitreous hemorrhage and enucleation was required.

Compared with eyes treated with plaque radiotherapy alone, the eyes treated with plaque radiotherapy plus adjuvant TTT showed less neovascular glaucoma ($P = .03$) and vitreous hemorrhage ($P = .049$) but greater papillopathy

Table 2. Characteristics of Choroidal Melanoma Overhanging the Optic Disc

Characteristic	Eyes, No. (%) (n = 141)
Tumor meridian	
Superior	33 (23)
Nasal	57 (40)
Inferior	38 (27)
Temporal	13 (9)
Proximity to foveola, median, mean (range), mm	2, 2 (0-5)
Tumor base, median, mean (range), mm	11, 11 (4-20)
Tumor thickness, median, mean (range), mm	5.2, 5.7 (1.3-14.8)
Subretinal fluid	124 (88)
Orange pigment	62 (44)
Tumor shape	
Plateau or dome	127 (91)
Mushroom	13 (9)
Color	
Pigmented	55 (39)
Partially pigmented	52 (37)
Nonpigmented	34 (24)
Growth pattern	
Circumscribed	128 (91)
Diffuse	13 (9)
Anterior limit of tumor	
Macula	4 (3)
Equator	97 (69)
Ora serrata	36 (26)
Ciliary body	4 (3)
Associated features	
Extraocular extension	3 (2)
Retinal invasion	22 (16)
Retinal vascular congestion	3 (2)
Overhang of optic nerve, %	
≤50	88 (62)
>50	53 (38)
Clock-hours tumor circumscribed optic nerve	
0-4	16 (11)
4-8	95 (67)
8-12	30 (21)
Optic disc swelling	5 (4)

($P = .01$) and all types of vascular occlusions. Outcomes were further evaluated for differences between the TTT group and the non-TTT group using the Cox proportional hazards model (eTable 3). After adjusting for baseline parameter differences, the only hazard ratios (HRs) reaching statistical significance were for papillopathy (HR, 2.06), major branch retinal artery occlusion (HR, 8.60), minor branch retinal vein occlusion (HR, 12.48), and major branch retinal vein occlusion (HR, 13.74). For all other outcomes, the difference between the 2 groups did not reach statistical significance.

Snellen visual acuity at last follow-up visit and number of eyes losing more than 5 Snellen lines were evaluated to assess long-term visual outcome. Final visual acuity of 20/200 or worse was observed in 72 eyes (77%). There were 41 eyes (89%) in the non-TTT group and 31 eyes (66%) in the TTT group that reached this level of vision or worse. Visual loss of more than 5 Snellen lines occurred in 59 eyes (63%), of which 30 were in the non-TTT group (65%) and 29 were in the TTT group (62%).

Enucleation was required in 27 eyes (23%), of which 20 eyes were in the non-TTT group (30%) and 7 were in the TTT group (13%). The reasons for enucleation were tumor recurrence in 7 eyes (26%), neovascular glaucoma in 14 eyes (52%), vitreous hemorrhage preventing adequate ophthalmoscopic monitoring of tumor status in 3 eyes (11%), and ocular pain in 3 eyes (11%).

During the median 46 months of follow-up, tumor recurrence was observed in a total of 12 eyes (10%), of which 5 eyes were in the TTT group (9%) and 7 eyes were in the non-TTT group (11%). Of the 12 recurrences, 5 had been treated with a round plaque (38% [n=13]), 6 had been treated with a notched plaque (6% [n=106]), and 1 had been treated with a rectangular plaque (100% [n=1]). Of the 12 eyes with recurrence, 10 showed 50% or less optic nerve overhang (13% [n=75]) and 2 showed greater than 50% overhang (4% [n=45]), though the difference between the overhang groups did not reach statistical significance. There were no cases of recurrence in eyes with 100% overhang (n=19). Recurrence was detected in 1 eye with less than 4 clock-hours' involvement (7% [n=15]), 10 eyes with 4 to 8 clock-hours' involvement (12% [n=81]), and 1 eye with greater than 8 clock-hours' involvement (4% [n=24]). Treatment for recurrence was enucleation (7 eyes), TTT (3 eyes), adjunctive argon laser photocoagulation (1 eye), and observation (1 eye).

There were 15 patients (13%) who had evidence of metastasis during the mean follow-up period of 56 months. In the non-TTT group, there were 10 cases of metastasis (15%; mean follow-up, 63 months), and in the TTT group, there were 5 cases (9%; mean follow-up, 46 months). Adjusting for differences in baseline parameters, there was no statistically significant difference in metastatic rates between the non-TTT and TTT groups. Death occurred in 4 cases (3%) overall. All of these patients were among the non-TTT group.

Table 5 lists outcomes in relation to extent of optic disc overhang. There was no statistically significant difference for any outcome between the 50% or less optic nerve overhang group and the greater than 50% overhang group.

COMMENT

In this article, we evaluate the clinical features and treatment outcomes of juxtapapillary choroidal melanoma in the special situation in which at least part of the optic disc is obscured by the tumor. Different therapies used to treat juxtapapillary melanoma, including enucleation,¹⁰ transpupillary thermotherapy,^{8,9} charged particle radiotherapy,⁶ stereotactic radiotherapy,⁷ and plaque radiotherapy,^{4,5} have been reported in the literature, but none of these articles have analyzed outcomes according to overhang of the disc by tumor. Previous studies have, however, demonstrated equivalence in survival of patients treated with plaque radiotherapy vs enucleation for juxtapapillary melanoma.^{5,14} The clinical finding of overhang of the optic disc by a tumor is a dramatic feature of some juxtapapillary melanomas. In a series of 650 eyes with juxtapapillary melanoma undergoing

Table 3. Plaque Radiotherapy Parameters in Patients With Choroidal Melanoma Overhanging the Optic Disc

Characteristic	Patients, No. (%)
Isotope (n = 141)	
Iodine 125	132 (94)
Cobalt 60	9 (6)
Plaque shape (n = 141)	
Round	14 (10)
Notched	126 (89)
Rectangular	1 (<1)
Distribution area (n = 137)	
Full	104 (76)
Posterior	33 (24)
Plaque size, median (range), mm (n = 140)	15 (12-22)
Radiation duration, median (range), h	98 (87-298)
Radiation dose, median (range), cGy	
At the apex	8500 (4500-15 500)
At the base	32 700 (11 000-59 600)
At the optic disc	12 034 (1373-87 773)
At the foveola	7935 (2380-51 158)
At the lens	1799 (490-9300)
Radiation rate, median (range), cGy/h	
At the apex	86 (23-325)
At the base	303 (77-589)
At the optic disc	119 (18-418)
At the foveola	87 (14-528)
At the lens	15 (5-641)

Table 4. Planned Adjunctive Treatment Following Primary Plaque Radiotherapy in Patients With Follow-up Data^a

Characteristic	Patients, No. (%) (n = 133)
Postbrachytherapy treatment	
ALPC	13 (10)
TTT	54 (41)
None	66 (50)
Total No. of TTT treatment sessions	
1	7 (13)
2	12 (22)
3	34 (63)
4	1 (2)
Location of TTT treatment	
Foveal	50 (93)
Extrafoveal	4 (7)

Abbreviations: ALPC, argon laser photocoagulation; TTT, transpupillary thermotherapy.

^aExcluded from this group are 3 patients who received TTT and 3 patients who received ALPC before requiring a plaque for recurrence.

plaque radiotherapy at the Oncology Service of Wills Eye Hospital, 22% of tumors overhung the optic disc.¹⁵ This group represents choroidal melanoma at a stage when the tumor has sufficient growth to overhang and obscure at least a portion of the disc.

The COMS did not evaluate radiotherapy for melanoma adjacent to the optic disc and offered enucleation instead.¹¹ Enucleation was an option for the cases presented herein, and the decision to use radiotherapy was based on the desire to preserve the eye owing to poor vision in the opposite eye, a small tumor, or patient preference. From the results of this study, we are in a better

Table 5. Summary of Outcomes Based on Extent of Disc Overhang

Outcome	Tumors, No. (%)			P Value ^a
	Overall (n = 120)	≤50% Overhang (n = 75)	>50% Overhang (n = 45)	
Complications				
Nonproliferative retinopathy	61 (51)	38 (51)	23 (51)	>.99
Proliferative retinopathy	26 (22)	12 (16)	14 (31)	.07
Maculopathy	44 (37)	26 (35)	18 (40)	.56
Papillopathy	57 (48)	36 (48)	21 (47)	>.99
Cataract ^b	66 (61)	43 (61)	23 (59)	.57
Neovascular glaucoma	24 (19)	13 (17)	11 (24)	.36
Vitreous hemorrhage	48 (40)	28 (37)	20 (44)	.45
Scleral necrosis	0	0	0	
Branch retinal artery occlusion				
Minor branch	4 (3)	3 (4)	1 (2)	>.99
Major branch	18 (15)	11 (15)	7 (16)	>.99
Branch retinal vein occlusion				
Minor branch	11 (9)	9 (12)	2 (4)	.21
Major branch	28 (23)	18 (24)	10 (22)	>.99
Central retinal artery or vein occlusion	0	0	0	
Visual acuity ^c				
Final ≤20/200	72 (77)	43 (75)	29 (80)	.56
Visual loss >5 Snellen lines	59 (63)	35 (61)	24 (67)	.57
Enucleation, all causes	27 (23)	18 (24)	9 (20)	.66
Tumor recurrence	12 (10)	10 (13)	2 (4)	.21
Metastasis	15 (13)	8 (11)	7 (16)	.57
Death	4 (3)	2 (3)	2 (4)	.63
Follow-up, mean, median (range), mo	56, 46 (3-249)	57, 45 (3-249)	53, 48 (4-234)	.89 ^d

^aFisher exact test.

^bExcludes previous pseudophakia and nonradiation cataract cases (5 in the ≤50% overhang group and 6 in the >50% overhang group).

^cExcludes enucleations (18 in the ≤50% overhang group and 9 in the >50% overhang group).

^dWilcoxon rank sum test.

position to counsel this patient group, as the effectiveness of plaque radiotherapy for melanoma overhanging the optic disc has not been previously reported.

Most patients had symptoms at initial examination (75%), mainly, decreased visual acuity (51%). Visual field defect was a feature in only 18 eyes (13%) despite all tumors demonstrating some overhang of the optic disc. In our series of patients, most tumors obscured less than 50% of the optic disc, but 13% demonstrated complete overhang. Although enucleation was a prominent option in these cases, our study showed that these eyes can be treated with plaque radiotherapy, signifying another option in addition to enucleation.

Custom-designed radioactive plaque therapy can deliver tumoricidal radiation to the entire tumor in an ellipse, hence sparing important structures.¹⁶ For melanoma overhanging the optic disc, the posterior placement of the radioactive seeds allowed for extension of the radiation field off the back margin of the plaque to encompass the eccentric tumor overhanging the disc. In addition, radioactive seeds positioned into the legs of the notch contribute to the posterior elliptical radiation field. It is important to know not only the tumor elevation and extent of disc overhang but also the extent of circumferential involvement in planning plaque radiotherapy. If the melanoma has less than 4 clock-hours of circumferential spread, then it is our preference to use a shallow-notched plaque, whereas greater circumferential involve-

ment requires a deep-notched plaque (most of our cases involved 4-8 clock-hours). The shape of the plaque was notched in 89% and round in 10%. Historically, all radioactive plaques were round; the use of round plaques for melanoma overhanging the optic disc dates back to the 1970s and early 1980s. Custom design of radiation field using notched plaques has been used for such tumors since the mid 1980s. In this analysis, recurrence of melanoma was 6 times more common following treatment with a round plaque compared with only 6% recurrence with a notched plaque. Notched plaque radiotherapy continues to be our preferred choice of plaque shape for juxtapapillary melanoma, particularly those that overhang the optic disc.

As plaque radiotherapy with adjuvant TTT is one of the current treatments for posterior uveal melanoma,¹⁷ we divided our series into those who were treated with plaque radiotherapy alone (66 eyes) and those who received adjuvant TTT (54 eyes). Entry characteristics were similar in the 2 groups except that larger tumor thickness and tumor base were found in the non-TTT group and greater proportion of pigmented tumors and those with orange pigment were included in the TTT group. Outcomes of treatment were assessed in the whole series and by TTT subgroup. Those eyes treated with plaque radiotherapy and TTT showed significantly more papillopathy, major branch retinal artery, and vein occlusions (Cox proportional hazards model).

Long-term visual outcomes were assessed by final visual acuity and by loss of vision of more than 5 Snellen lines. Overall, 77% of treated eyes had a final visual acuity of 20/200 or worse, with no significant difference between the non-TTT group (89%) and the TTT group (66%). Visual loss of more than 5 Snellen lines occurred in 63% of eyes, with equal proportions in the 2 subgroups. Visual loss is known to be related to the juxtapapillary location of the tumor as a result of radiation complications and the effects of subfoveal tumor with related subfoveal fluid. In an analysis of visual outcome of 1106 consecutive patients with choroidal melanoma in all locations, proximity of the tumor to the foveola and optic disc were substantial risk factors for visual loss.¹⁸ In a previous series of juxtapapillary melanoma treated with plaque radiotherapy, vision loss of at least 3 Snellen acuity lines was found in 72% of eyes at 60 months.¹⁴

Enucleation was eventually necessary in 23% of patients (30% in the non-TTT group compared with 13% in the TTT group). This observed difference in the enucleation rate between the 2 groups did not reach statistical significance. The most common reason for enucleation was neovascular glaucoma (52%), though tumor recurrence (26%), ocular pain (11%), and vitreous hemorrhage (11%) were other causes. Enucleation for vitreous hemorrhage is only necessary when poor fundus view makes monitoring of the tumor unsatisfactory, especially if there is any suspicion of recurrence. Following conservative treatment of uveal melanoma with plaque radiotherapy, proton beam radiotherapy, or helium ion radiotherapy, enucleation rates have ranged from 6% to 40%.^{6,19-22} Reported factors related to the need for enucleation following radiotherapy include greater tumor thickness, closer proximity of the posterior tumor border to the foveal avascular zone, poorer baseline visual acuity, tumor recurrence, neovascular glaucoma, sclerocorneal melt, painful bullous keratopathy, and occasionally patient request.^{21,23,24} In a previous report of juxtapapillary melanoma treated with plaque radiotherapy,¹⁴ the enucleation rate was 22%, which is similar to the findings of the current study.

Melanoma recurrence was found in 10% of patients, with similar rates among the non-TTT and TTT groups. Notched radioactive plaques showed less recurrence (6%) relative to round plaques (38%). We analyzed recurrence rates by extent of optic disc overhang to determine whether there was a pattern. There was no statistically significant difference for recurrence, or for any outcome, between the 50% or less optic nerve overhang group and the greater than 50% overhang group. There were no cases of recurrence in eyes with 100% overhang of the optic disc, suggesting that this advanced feature should not rule out the use of plaque radiotherapy.

Metastasis was noted in 13% of patients overall, with 15% in the non-TTT group and 9% in the TTT group. Death occurred in 3% of patients during the study period. All of these eyes were treated with plaque alone and may be a reflection of the longer follow-up in the non-TTT group. It is entirely possible that some melanoma deaths occurred after the study period or were not reported to us.

We conclude that juxtapapillary melanoma that overhangs the optic disc can be treated with plaque radiotherapy with local tumor control in 90% of eyes. Metastatic rate was 13% and mortality rate was 3%. The feature of the tumor overhanging the optic disc should not preclude the use of this treatment, which can be considered an alternative to enucleation in selected cases.

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Additional Information: The eTables are available at <http://www.archophthalmol.com>.

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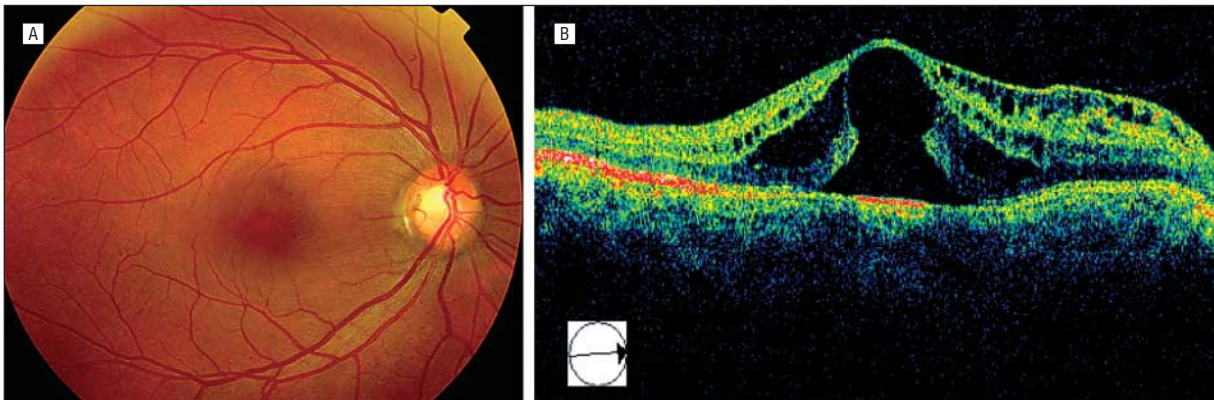
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Ophthalmic Images

Optic Disc Pit and Outer Lamellar Macular Hole

Mor Ndiaye; Marie-Noelle Delyfer, MD; Marie-Benedicte Rougier, MD; Jean-Francois Korobelnik, MD



A 35-year-old woman had decreased vision related to a temporal optic disc pit. Optical coherence tomography (Stratus OCT 3, Zeiss, Jena, Germany) showed a schisis of the retina connected to the pit, cystic cavities in the inner layers of the retina, and an outer macular hole, without visible traction of the vitreous.